

Reducing Cardiovascular Disease in Indonesia

Health Impact, Costs, and Economic Returns of Clinical and Public-Health Interventions, 2025–2050

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Reducing CVD in Indonesia

Why this matters for Indonesia

Cardiovascular disease is Indonesia's largest health burden

Under current trends, a person reaching age 30 faces roughly a **14.5% chance** of dying from cardiovascular disease before age 70 — the CVD “40q30”. Most of this is **preventable or treatable**.

- Premature CVD deaths (ages 30–69) have risen for two decades and remain high.
- Two complementary levers:
 - **Clinical care** — prevent and better treat established disease.
 - **Population prevention** — tobacco, salt, trans-fat, sugary drinks, alcohol.
- This analysis weighs their **health impact, cost, and economic return** together.

Source: Indonesia CVD model (calibrated to GBD 2023; UN World Population Prospects 2024). CVD 40q30 = probability of dying from cardiovascular disease between exact ages 30 and 70.

What this analysis set out to do

General aim

Estimate the **health and economic effects** of scaling up evidence-based **clinical** and **public-health** cardiovascular interventions in Indonesia through **2050**.

Aim 1 — Burden

Project the CVD burden and the premature-mortality indicator (CVD 40q30) under the **status quo** and under **alternative intervention scenarios**.

Aim 2 — Value for money

Estimate **deaths averted**, mortality reductions, **incremental costs**, and **cost-effectiveness** for each intervention and package.

Aim 3 — Economic return

Estimate **economic benefits**, **benefit-cost ratios**, and the **added value of combining** clinical and public-health action.

Horizon 2025–2050; the “status quo” baseline is the no-new-intervention comparator against which every scenario is measured.

Executive summary — the headline numbers

Combining clinical + public health is the big prize

Together they could avert about **3.5M deaths** over 2025–2050, cut the CVD 40q30 in 2050 by **29%**, and return about **79.1 int\$** in benefits per int\$ spent.

- **All clinical interventions:** $\approx 2.0\text{M}$ deaths averted; CVD 40q30 down 20% by 2050.
- **All public-health policies:** $\approx 1.6\text{M}$ deaths averted; benefit–cost ratio $\approx 1,240$.
- **Most impactful single levers:** **Primary CVD prevention** and **Tobacco tax**.
- **Affordability:** the combined package costs about US\$4,814 per death averted.

All figures from the model's cost/value workbooks. Deaths averted are cumulative 2025–2050 across modeled cardiovascular and diabetes causes; benefits valued per the Reference Case (Robinson 2019) in PPP international dollars (Robinson, Hammitt, and O'Keeffe 2019).

How the model works — in plain language

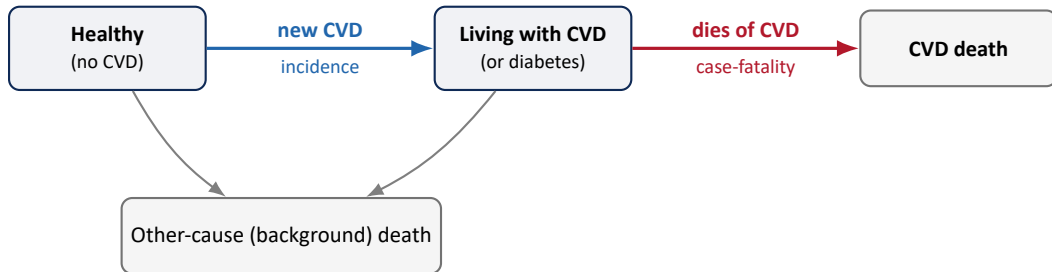
1. **People & demography.** Indonesia's population is projected by age and sex (UN World Population Prospects 2024).
2. **Starting health.** Disease and death rates are set to Indonesia's data (GBD 2023) and calibrated to match observed levels.
3. **Year-by-year simulation.** Each year people can develop cardiovascular disease or diabetes, survive, or die.
4. **Interventions.** Clinical and public-health scenarios change how many people develop disease and how many survive it.
5. **Outcomes.** The model reports deaths, healthy years (DALYs), premature-mortality (40q30), costs, cost-effectiveness, and economic returns.

One consistent engine

Every scenario runs through the **same** projection, so results are directly comparable and the **combined** package is modeled as one run — not a sum of separate estimates.

Calibration fits the model to GBD estimates using a standard numerical optimiser (Nelder–Mead). Life expectancy for lost-years and economic value uses UN WPP 2024.

How interventions change the disease pathway



Fewer people fall ill

Primary prevention & *all* public-health policies lower **incidence**.

More people survive

Secondary prevention, treatment & tobacco policy lower **case-fatality**.

Background mortality

Other-cause deaths from demography; unchanged by these interventions.

No recovery/remission transition is modeled: people living with CVD remain at elevated risk (a conservative assumption).

Source: model engine, 06_run_scenarios_indonesia_fair.R.

The interventions

Clinical interventions (FAIR Choices)

Intervention	Target population	Model effect
Primary CVD prevention	Adults at cardiovascular risk	Incidence
Type-2 diabetes treatment	People with type-2 diabetes	Case-fatality
Heart-attack secondary prevention	Heart-attack (IHD) survivors	Case-fatality
Stroke secondary prevention	Stroke survivors	Case-fatality
Chronic & acute heart-failure care	Heart-failure patients	Case-fatality
RHD secondary prevention & surgery	Rheumatic heart disease patients	Case-fatality

Implementation scenario (all clinical interventions): coverage scales up **linearly from 31% to 80%** of the eligible population between 2025 and 2050 (RHD surgery from a lower base). Effect sizes are drawn from the FAIR Choices intervention taxonomy; primary prevention lowers who develops CVD, while secondary prevention and treatment improve survival. Source: workbook `Selected_Interventions`.

Public-health interventions (fiscal & regulatory policy)

Policy	Risk factor	Model effect	Implementation lever
Tobacco tax	Tobacco	Incidence + survival	Raise tax to 75% of retail price
Tobacco policy package	Tobacco	Incidence + survival	Clean-air law, mass media, ad ban
Sodium policy package	Dietary salt/sodium	Incidence	Reformulation, labelling, media, procurement
Trans-fat elimination	Industrial trans-fat	Incidence	Best-practice industrial TFA ban
Sugary-drink (SSB) tax	Sugary drinks	Incidence	Price increase via excise
Alcohol tax / ad ban	Alcohol	Incidence	Price increase; advertising ban

Policies reduce exposure (smoking, salt, trans-fat, sugary-drink and alcohol intake), lowering CVD **incidence**. **Tobacco** policies additionally improve **survival** after CVD. Packages are modeled as single joint runs of their component measures. Source: workbook `Selected_Interventions / Policy_Lever`s.

Data sources

Model component	Primary source	Years / version	Use in model
Population & demography	UN World Population Prospects	2024 revision	Population, background mortality, life expectancy
Cause-specific mortality & disease rates	GBD (Indonesia)	2023	Baseline burden & calibration targets
Disability weights (healthy-years lost)	GBD	2023	DALYs
Clinical intervention effects	FAIR Choices intervention taxonomy	current	Incidence & case-fatality effects
Public-health effect sizes	Published risk-factor evidence (in workbook)	current	Incidence & survival effects
Intervention & policy costs	Programme & policy unit costs (in workbook)	price year 2023	Budget impact & cost-effectiveness
Economic valuation (VSL/VSLY)	Reference Case guidance (Robinson 2019)	2019	Monetized benefits & benefit–cost
International 40q30 comparison	WPP-scaled ASEAN comparison series	1995–2050	Benchmarking Indonesia vs. neighbour

Sources verified in the modeling pipeline and input workbooks. Specific effect-size and cost citations are documented in the input workbooks. Population: UN World Population Prospects 2024; GBD (GBD 2023 Cardiovascular Diseases and Risks Collaborators 2025; Schumacher et al. 2024); economic valuation (Robinson, Hammitt, and O’Keeffe 2019).

Health impact and value for money

Public health: impact by policy

Public-health policy	Deaths averted (cumulative)	CVD 40q30 2050	40q30 reduction 2050	Deaths reduction 2050
Tobacco tax	724,000	8.6%	5.2%	3.3%
Sodium policy package	540,000	8.7%	4.0%	3.0%
Tobacco policy package	435,000	8.8%	3.1%	2.0%
Mandatory sodium reformulation	294,000	8.9%	2.2%	1.6%
Tobacco mass-media campaign	178,000	9.0%	1.3%	0.8%
Smoke-free / clean-air law	150,000	9.0%	1.1%	0.7%
Sodium front-of-pack labelling	148,000	9.0%	1.1%	0.8%
Tobacco advertising ban	109,000	9.0%	0.8%	0.5%
Supportive public-institution environment	104,000	9.0%	0.8%	0.6%
Sodium mass-media campaign	0	9.1%	-0.0%	-0.0%
All public health (combined)	1,649,000	8.0%	11.9%	8.0%

Cumulative deaths averted over 2025–2050 vs. the status-quo baseline. Packages (tobacco, sodium) are single joint runs of their component measures; individual sodium/tobacco measures are listed in the appendix. Source: public-health cost/value workbook.

Clinical: impact and cost-effectiveness

Clinical intervention	Deaths averted (cumulative)	CVD 40q30 2050	40q30 red. 2050	Deaths red. 2050	Cost per death averted	Annual cost per capita
Primary CVD prevention	1,322,000	7.6%	16.5%	10.9%	US\$919	US\$0.25
Type-2 diabetes treatment	369,000	9.1%	0.3%	2.4%	US\$26,411	US\$1.97
Heart-attack secondary prevention	141,000	8.9%	1.8%	0.8%	US\$21,376	US\$0.61
Chronic heart-failure care	77,000	9.0%	0.9%	0.5%	US\$7,831	US\$0.13
Acute heart-failure care	69,000	9.0%	0.6%	0.5%	US\$13,210	US\$0.19
Stroke secondary prevention	31,000	9.1%	0.3%	0.2%	US\$56,216	US\$0.36
RHD secondary prevention	6,000	9.1%	0.0%	0.0%	US\$24,755	US\$0.03
All clinical (combined)	1,978,000	7.3%	19.8%	14.9%	US\$8,312	US\$3.33

Cost per death averted uses **discounted** incremental cost (3%/yr). Annual cost per capita = undiscounted incremental cost averaged over 26 years and Indonesia's population. Costs in market US\$ (price year 2023); perspective: health system. Source: clinical cost/value workbook.

Clinical: benefit–cost analysis

Clinical intervention	Deaths averted	Cost per death averted	Benefits (PV)	Net benefit (PV)	Benefit–cost ratio
Primary CVD prevention	1,322,000	US\$919	Int\$1.5T	Int\$1.5T	418
Type-2 diabetes treatment	369,000	US\$26,411	Int\$421.3B	Int\$392.0B	14.4
Heart-attack secondary prevention	141,000	US\$21,376	Int\$159.9B	Int\$150.8B	17.7
Chronic heart-failure care	77,000	US\$7,831	Int\$87.7B	Int\$85.9B	48.4
Acute heart-failure care	69,000	US\$13,210	Int\$78.2B	Int\$75.5B	28.8
Stroke secondary prevention	31,000	US\$56,216	Int\$35.7B	Int\$30.4B	6.7
RHD secondary prevention	6,000	US\$24,755	Int\$6.9B	Int\$6.5B	15.3
All clinical (combined)	1,978,000	US\$8,312	Int\$2.3T	Int\$2.2T	46.0

Benefits = value of averted mortality (preferred VSL, income elasticity 1.5), in PPP international dollars, base year 2025, discounted at 3%/yr; standing: societal; scope: partial (mortality benefit only). A ratio above 1 means benefits exceed costs. Source: clinical cost/value workbook, preferred-VSL case.

Public health: cost-effectiveness

Public-health policy	Deaths averted (cumulative)	CVD 40q30 2050	40q30 red. 2050	Deaths red. 2050	Cost per death averted	Annual cost per capita
Tobacco tax	724,000	8.6%	5.2%	3.3%	US\$283	US\$0.04
Sodium policy package	540,000	8.7%	4.0%	3.0%	US\$333	US\$0.03
Tobacco policy package	435,000	8.8%	3.1%	2.0%	US\$272	US\$0.02
Mandatory sodium reformulation	294,000	8.9%	2.2%	1.6%	US\$153	US\$0.01
Tobacco mass-media campaign	178,000	9.0%	1.3%	0.8%	US\$222	US\$0.01
Smoke-free / clean-air law	150,000	9.0%	1.1%	0.7%	US\$262	US\$0.01
Sodium front-of-pack labelling	148,000	9.0%	1.1%	0.8%	US\$303	US\$0.01
Tobacco advertising ban	109,000	9.0%	0.8%	0.5%	US\$361	US\$0.01
Supportive public-institution environment	104,000	9.0%	0.8%	0.6%	US\$433	US\$0.01
Sodium mass-media campaign	0	9.1%	-0.0%	-0.0%	US\$NA	US\$0.01
All public health (combined)	1,649,000	8.0%	11.9%	8.0%	US\$305	US\$0.09

Fiscal and regulatory policies reach the whole population at very low per-person cost, so cost per death averted is far lower than for clinical care. Same costing conventions as the clinical table. Source: public-health cost/value workbook.

Public health: benefit–cost analysis

Public-health policy	Deaths averted	Cost per death averted	Benefits (PV)	Net benefit (PV)	Benefit–cost ratio
Tobacco tax	724,000	US\$283	Int\$819.8B	Int\$819.2B	1,330
Sodium policy package	540,000	US\$333	Int\$616.8B	Int\$616.3B	1,140
Tobacco policy package	435,000	US\$272	Int\$492.0B	Int\$491.6B	1,390
Mandatory sodium reformulation	294,000	US\$153	Int\$336.5B	Int\$336.3B	2,500
Tobacco mass-media campaign	178,000	US\$222	Int\$201.2B	Int\$201.1B	1,700
Smoke-free / clean-air law	150,000	US\$262	Int\$170.1B	Int\$170.0B	1,440
Sodium front-of-pack labelling	148,000	US\$303	Int\$169.1B	Int\$169.0B	1,260
Tobacco advertising ban	109,000	US\$361	Int\$123.6B	Int\$123.5B	1,050
Supportive public-institution environment	104,000	US\$433	Int\$118.6B	Int\$118.5B	880
Sodium mass-media campaign	0	US\$NA	Int\$0.0	Int\$-134.7M	-0.0
All public health (combined)	1,649,000	US\$305	Int\$1.9T	Int\$1.9T	1,240

Because fiscal/regulatory policies cost little to implement relative to the mortality they avert, their benefit–cost ratios are very large. Same valuation conventions (preferred VSL, PPP international dollars, base year 2025, 3%/yr) as the clinical BCA. Source: public-health cost/value workbook.

The combined package

Clinical + public health: better together

Package	Deaths averted (cumulative)	40q30 red. 2050	CVD 40q30 2050	Total cost (PV)	Cost per death averted	Benefit–cost ratio
All clinical (combined)	1,978,000	19.8%	7.3%	US\$16.4B	US\$8,312	46.0
All public health (combined)	1,649,000	11.9%	8.0%	US\$503.2M	US\$305	1,240
Clinical + public health (combined)	3,453,000	29.2%	6.4%	US\$16.6B	US\$4,814	79.1

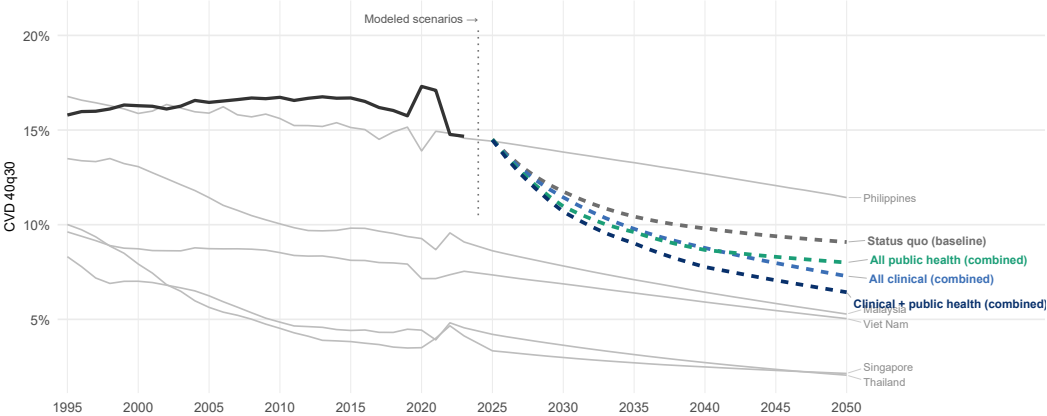
The combined package is modeled as one run — not a sum

Because clinical and public-health effects overlap, the joint scenario is projected in a single run. It averts about **3.5M deaths**, brings the 2050 CVD 40q30 down to **6.4%** (from 9.1% at baseline), and yields a benefit–cost ratio of about **79.1** — a net benefit of roughly **Int\$3.9T**.

CVD premature mortality: trajectory and neighbours

CVD 40q30 in Indonesia: observed, projected, and neighbours

Probability of dying from CVD between ages 30 and 70



Indonesia observed history (solid, to 2023) and ASEAN comparators are WPP-scaled reference series.
Modeled Indonesia scenarios (dashed, 2025-2050) use the GBD-calibrated model baseline, which sits somewhat higher.

What it means for Indonesia

Policy implications

- **Biggest health gains come from doing both.** The combined package averts far more (~3.5M) than either family alone.
- **Public-health policies are exceptional value.** Population-wide fiscal/regulatory measures avert deaths at a fraction of clinical cost (ratios in the thousands).
- **Clinical scale-up delivers large, durable gains** in survival and premature mortality, at conventional cost-effectiveness.
- **Sequence for early wins.** Tobacco taxation and other prevention policies can start cutting risk quickly and cheaply, while clinical capacity scales up.
- **The 40q30 target is within reach.** The combined package bends Indonesia's premature-CVD-mortality curve down by about 29% by 2050.

These are model findings; translating them into policy requires judgement on feasibility, financing and delivery. Model results quantify the *potential*; they are not a delivery guarantee.

Limitations

- Results depend on **modeled epidemiological estimates** (GBD) and calibration; real burden may differ.
- **Effect sizes, future coverage, and unit costs are uncertain**; interventions are assumed to reach and sustain their targets.
- The disease model is **simplified** (healthy → living-with-CVD → death; no remission; no comorbidity interactions).
- Economic returns value **averted mortality only** (a partial benefit-cost scope) — morbidity, productivity and downstream cost offsets are omitted, so returns are, if anything, understated.
- Projections are **scenarios, not forecasts or guarantees**; they show what is achievable if interventions are delivered as modeled.

Three things to remember

1. Both, together

Clinical + public health averts **~3.5M deaths** by 2050 — more than either alone.

2. Prevention pays

Population policies return on the order of **1,240×** their cost.

3. A reachable target

Premature CVD mortality (40q30) falls **29%** by 2050 under the combined package.

Combined action against CVD could avert about **3.5M deaths** and return about **79.1 int\$ per int\$ invested** through 2050.

Technical appendix